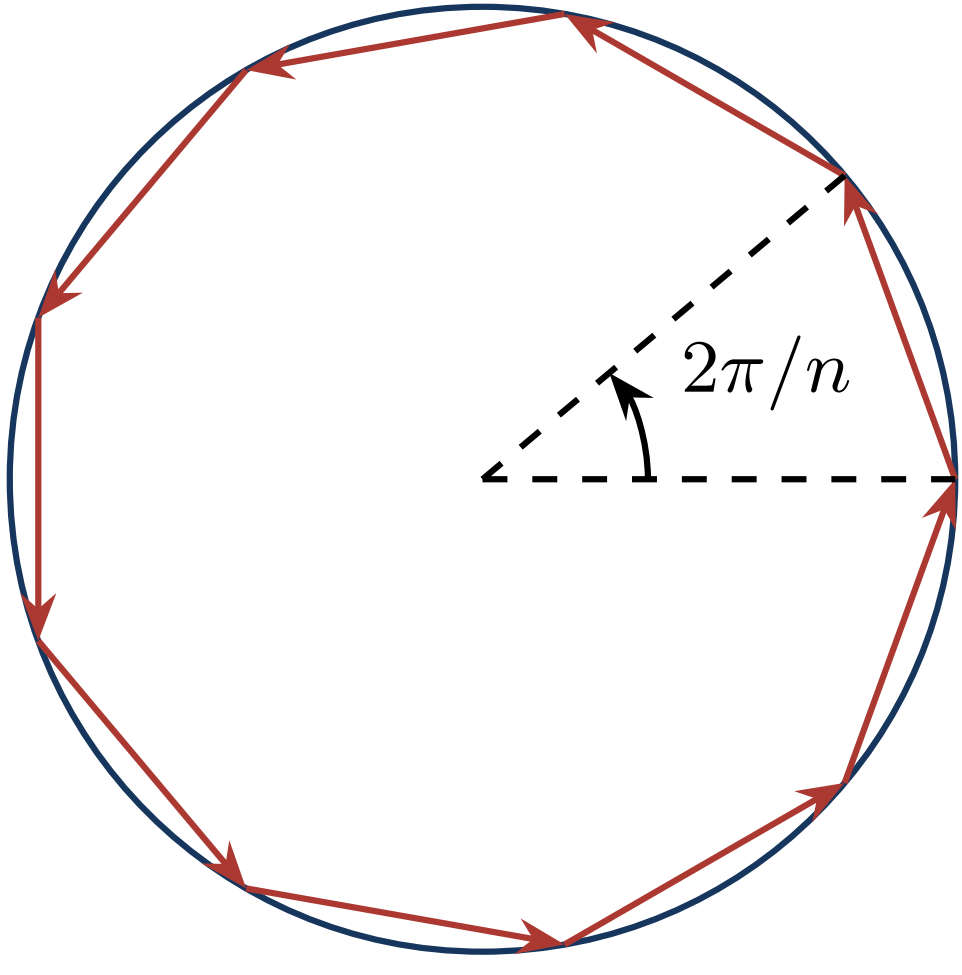
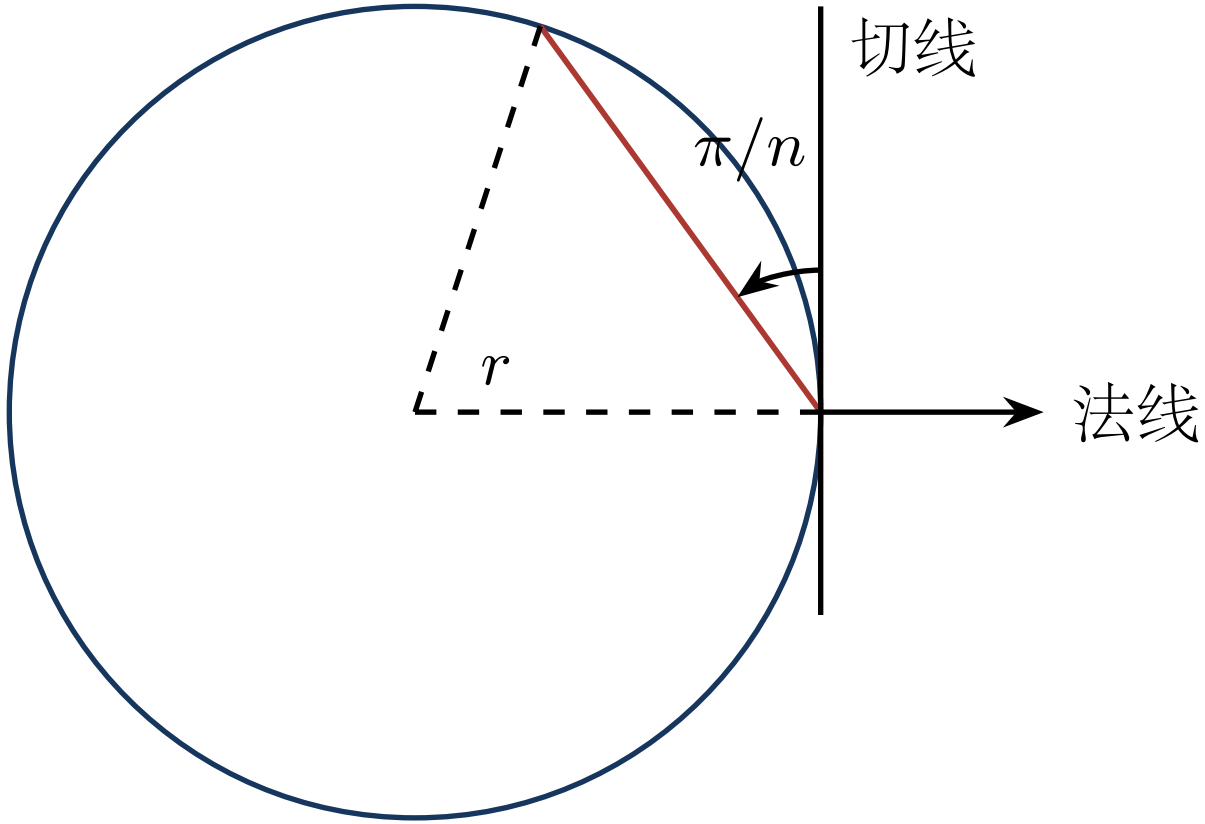
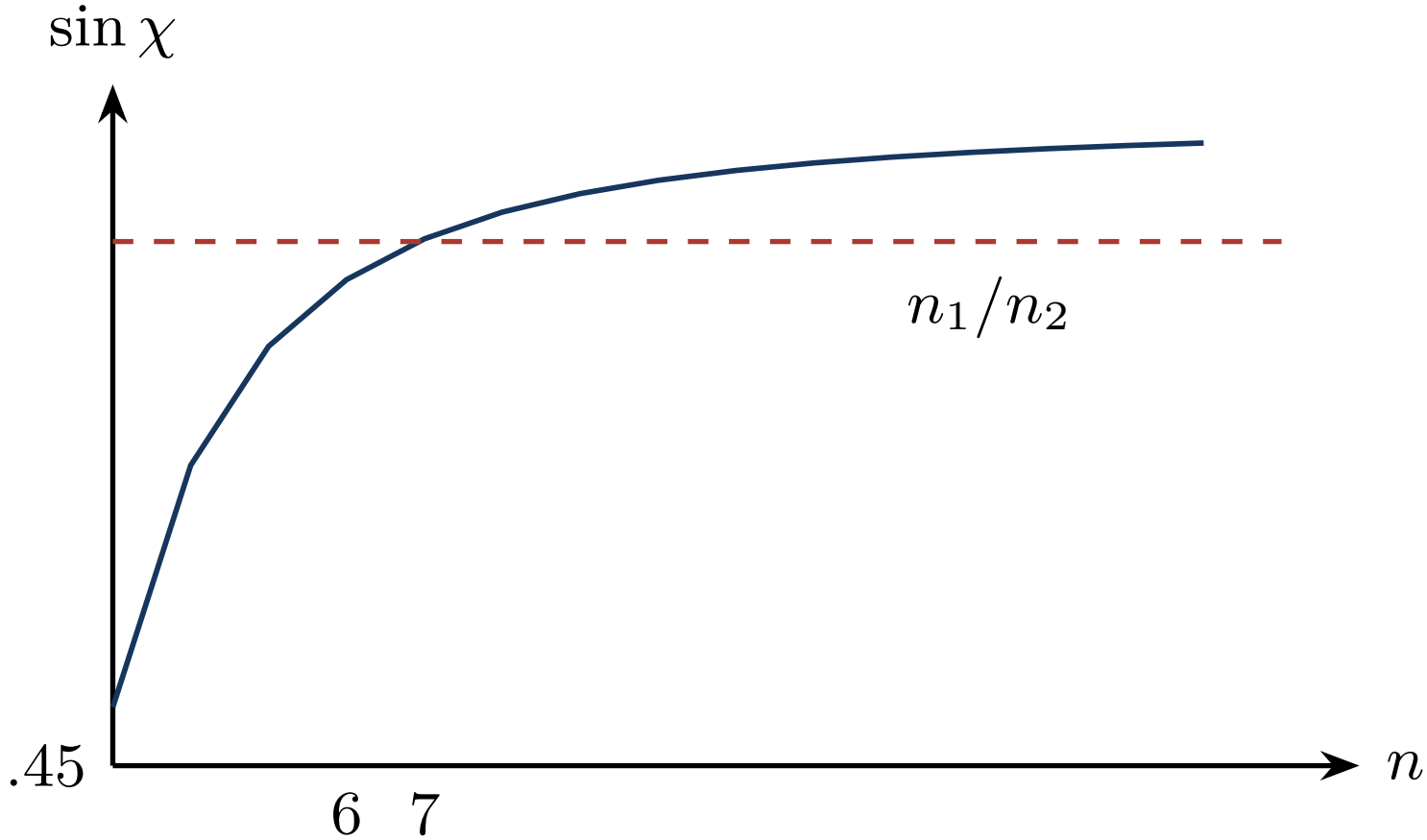


一圈闭合：正 n 边形，图示 $n = 9$



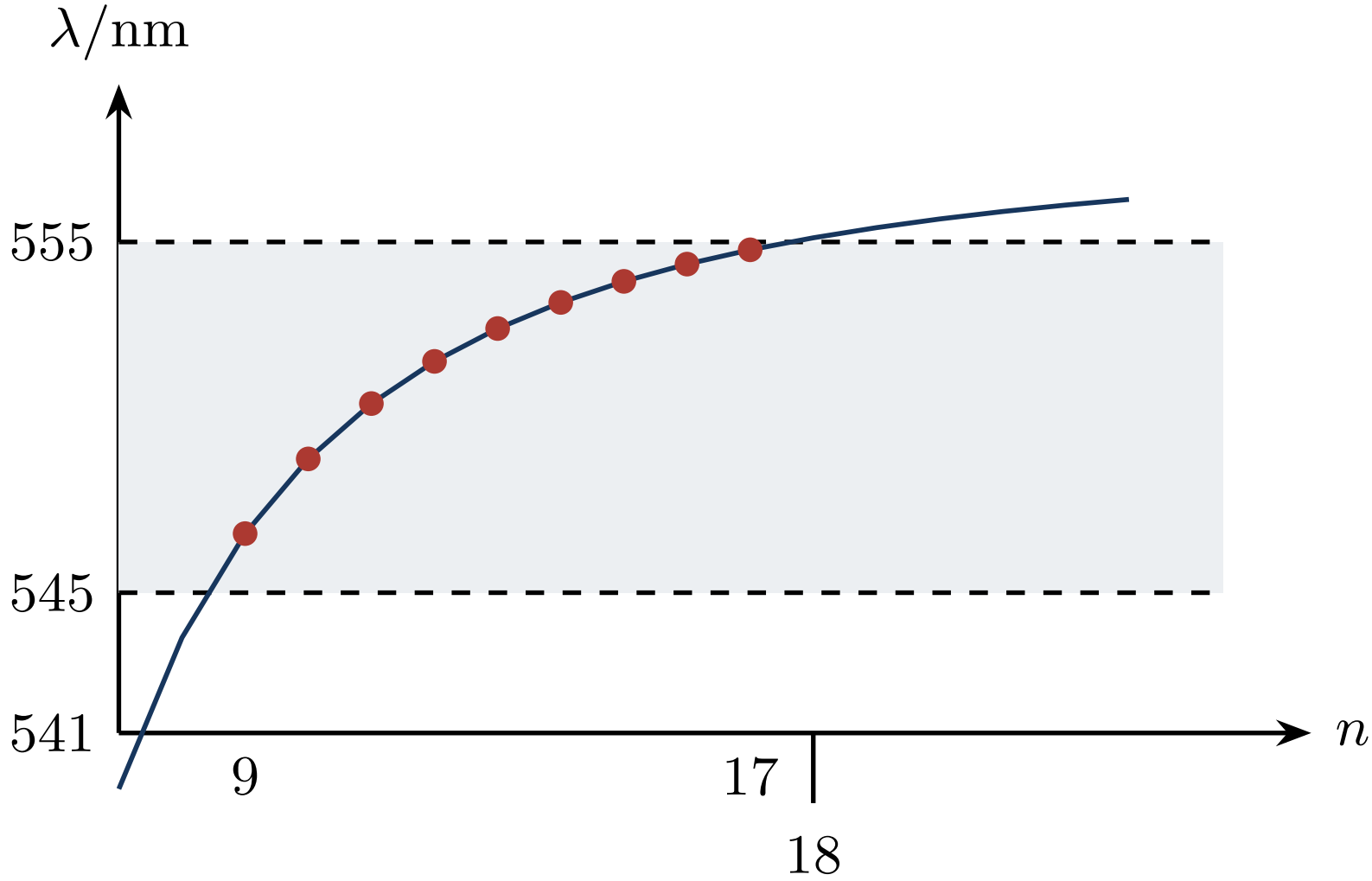
弦长 $2r\sin(\pi/n)$

法线入射角 $\chi = \pi/2 - \pi/n$

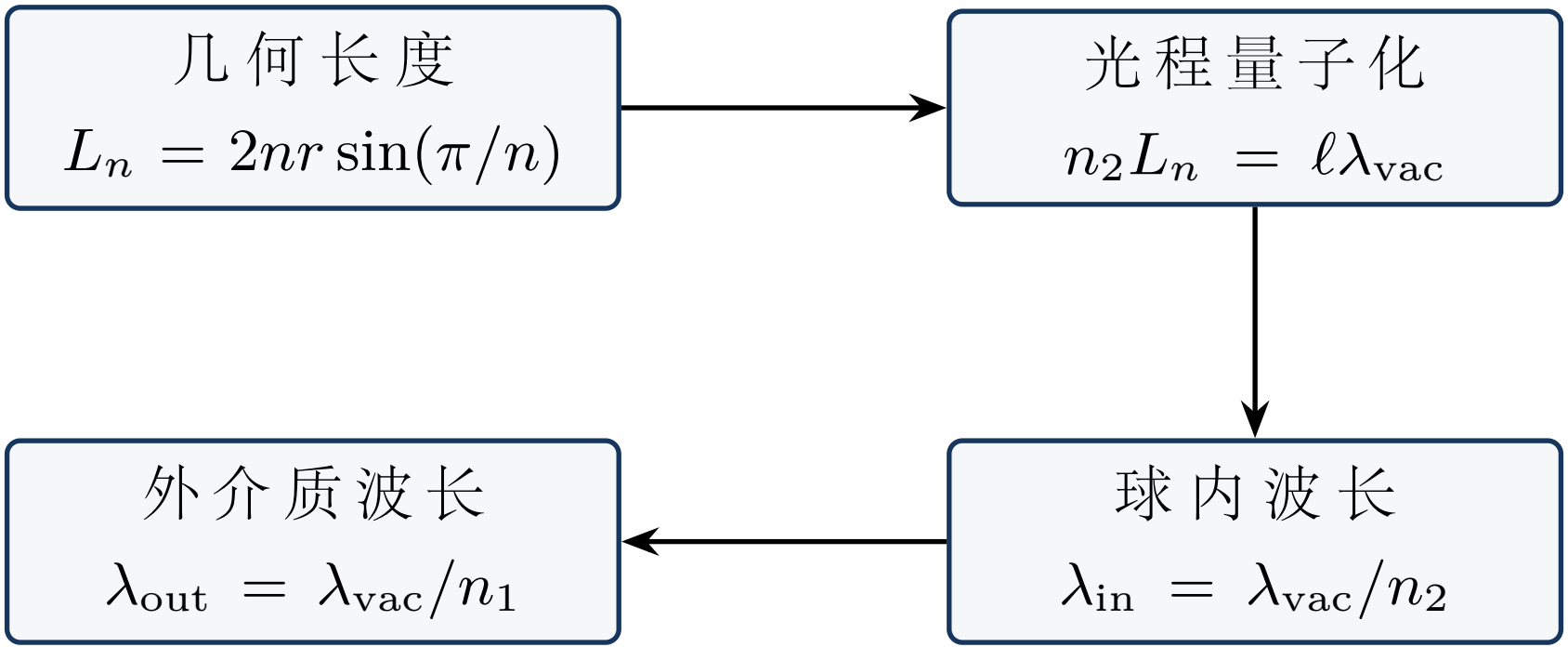


$$\cos(\pi/6) \quad < \quad n_1/n_2 \quad < \quad \cos(\pi/7)$$

最小全反射次数 $n \quad = \quad 7$



$\ell = 2$，允许 $n = 9, 10, \ldots, 17$



表中用真空波长；忽略每次全反射的相移